

PYTHON CW(4/8/26)

1.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

 **Sivarangjenii B**
192521375RD

```
1 data = set()
2
3 while 1:
4     ch = int(input())
5
6     if ch == 1:
7         n = input()
8         data.add(n)
9         print("Student added successfully.")
10
11    elif ch == 2:
12        n = input()
13        if n in data:
14            data.discard(n)
15            print("Student removed successfully.")
16        else:
17            print("Student not found.")
18
19    elif ch == 3:
20        n = input()
21        if n in data:
22            print("Student found.")
23        else:
24            print("Student not found.")
25
26    elif ch == 4:
27        print("Students:", data)
28
29    elif ch == 5:
30        print("Total number of students:", len(data))
31
32    elif ch == 6:
33        print("Program terminated.")
34        break
```

1
Ravi
1
Priya

Output
Student added successfully.
Student added successfully.
Students: {'Priya', 'Ravi'}
Student found.
Student removed successfully.
Total number of students: 1

2.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

 **Sivarangjenii B**
192521375RD

Symmetric Difference: {1, 2, 5, 6}

Sample Input

Enter your choice: 5

Sample Output

Program terminated.

Your Code

```
1 a = set(map(int, input().split()))
2 b = set(map(int, input().split()))
3 ch = int(input())
4
5 if ch == 1:
6     print("Union:", a | b)
7
8 elif ch == 2:
9     print("Intersection:", a & b)
10
11 elif ch == 3:
12     print("Difference:", a - b)
13
14 elif ch == 4:
15     print("Symmetric Difference:", a ^ b)
16
17 elif ch == 5:
18     print("Program terminated.")
```

Submitted

Input

```
1 2 3 4
3 4 5 6
1
```

Output

```
Union: {1, 2, 3, 4, 5, 6}
```

3.

SIMATS Online Programming Lab

Sivarangeni B
192521375RD

PYTHON PROGRAMMING

Q3 Exactly One Set

5 marks

Write a Python program to find the elements that appear in exactly one of three sets.

Input:

A={1,2,3}

B={3,4,5}

C={5,6,7}

Output:

{1,2,4,6,7}

Your Code

```
1 A = {1, 2, 3}
2 B = {3, 4, 5}
3 C = {5, 6, 7}
4
5 only_A = A - (B | C)
6 only_B = B - (A | C)
7 only_C = C - (A | B)
8
9 result = sorted(only_A | only_B | only_C)
10
11 print("{ " + ",".join(map(str, result)) + "}")
```

Input

stdin input

Output

{1, 2, 4, 6, 7}

Visible Test Cases

Test Case 1

5.00 / 5 marks

4.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B

192521375RD

← Back

Q4 Unique Words

5 marks

Write a Python program to display the unique words in a sentence using a set.

Input:
python is easy python
{'python','is','easy'}

Your Code

```
1 a=input().split()
2 res=[]
3 for i in a:
4     if i not in res:
5         res.append(i)
6 print ("{'"+ "','".join(res)+"'}")
```

Input

python is easy python

Output

```
{'python','is','easy'}
```

Visible Test Cases

Test Case 1

5.00 / 5 marks

Submitted

5.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B

192521375RD

← Back

Q5 Unique Characters

5 marks

Write a Python program to count the number of unique characters in a string.
Input:
Programming
output:
8

Your Code

```
1 a = input()  
2 print(len(set(a)))
```

Input

Programming

Output

8

Visible Test Cases

Test Case 1

✓

5.00 / 5 marks

✓ Submitted

6.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

 Sivarangjenii B
192521375RD

Q6 Pair Sum

5 marks

Write a Python program to find all pairs of elements whose sum equals a given number.

Input:

List: 2 4 6 8 10

Target: 12

Output:

(4,8),(2,10)

Your Code

```
1 lst = [2, 4, 6, 8, 10]
2 target = 12
3
4 pairs = []
5
6 for i in range(len(lst)):
7     for j in range(i + 1, len(lst)):
8         if lst[i] + lst[j] == target:
9             pairs.append((lst[i], lst[j]))
10
11 pairs.sort(reverse=True)
12
13 for i in range(len(pairs)):
14     print(f"({pairs[i][0]}, {pairs[i][1]})", end=" ")
15     if i != len(pairs) - 1:
16         print(", ", end="")
```

Input

2 4 6 8 10
12

Output

(4,8), (2,10)

Visible Test Cases

Test Case 1

5.00 / 5 marks

Submitted

7.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

Q7 Duplicate Tuples

5 marks

Write a Python program to remove duplicate tuples from a list using a set.

Input:

```
[(1,2),(1,2),(3,4)]
```

Output:

```
[(1,2),(3,4)]
```

Your Code

```
1 l=[(1,2),(1,2),(3,4)]
2 s=sorted(set(l))
3 print(" "+".join(["("+str(i[0])+","+str(i[1]))" for i in s])+"")
```

Input

stdin input

Output

```
[(1,2),(3,4)]
```

Visible Test Cases

Test Case 1

5.00 / 5 marks

Submitted

8.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjenii B
192521375RD

Q8 Compare Lists

5 marks

Write a Python program to check whether two lists contain the same unique elements.

Input:

List1:1 2 2 3

List2:3 2 1

Output:

True

Your Code

```
1 list1 = list(map(int,input().split()))
2 list2 = list(map(int,input().split()))
3 if set(list1) == set(list2):
4     print(True)
5 else:
6     print(False)
```

Input

```
1 2 2 3
3 2 1
```

Output

```
True
```

Visible Test Cases

Test Case 1

0.00 / 5 marks

Run

Save

Submit

10.

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Sivarangjeni B
192521375RD

Back

Q10 First Repeated

5 marks

Write a Python program to find the first repeated element in a list using a set.

Input:

10 20 30 20 40

output:

20

Your Code

```
1 l=list(map(int,input().split()))
2 seen=set()
3 for i in l:
4     if i in seen:
5         print(i)
6         break
7     seen.add(i)
```

Input

10 20 30 20 40

Output

20

Visible Test Cases

Test Case 1

5.00 / 5 marks

Submitted